In [2]:
```
import pandas as pd

df = pd.read_csv("customer_shopping_behavior.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Disc App
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	

In [ ]:

Checking the Nulls and data types of columns

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [ ]:

Summary statistics of numerical columns

In [5]:
```
df.describe()
```

Out[5]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

In [16]:

```
df.columns
```

Out[16]:

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

In [ ]:

Maintaining readability in column names

In [13]:

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(" ","_")
```

In [ ]:

Checking nulls

In [17]:
```
df.isna().sum()
```

Out[17]:
```
customer_id               0
age                       0
gender                    0
item_purchased            0
category                  0
purchase_amount_(usd)     0
location                  0
size                      0
color                     0
season                    0
review_rating            37
subscription_status       0
shipping_type             0
discount_applied          0
promo_code_used           0
previous_purchases        0
payment_method            0
frequency_of_purchases    0
dtype: int64
```

In [19]:
```
df.groupby(['category'])['review_rating'].mean()
```

Out[19]:
```
category
Accessories    3.769976
Clothing       3.721537
Footwear       3.793771
Outerwear      3.745652
Name: review_rating, dtype: float64
```

In [ ]:

Filling Nulls

In [22]:
```
df['review_rating'] = df['review_rating'].fillna(df.groupby('category')['review_rating'].transform('median'))
```

In [23]:
```
df.isna().sum()
```

Out[23]:
```
customer_id               0
age                       0
gender                    0
item_purchased            0
category                  0
purchase_amount_(usd)     0
location                  0
size                      0
color                     0
season                    0
review_rating             0
subscription_status       0
shipping_type             0
discount_applied          0
promo_code_used           0
previous_purchases        0
payment_method            0
frequency_of_purchases    0
dtype: int64
```

In [ ]:

Creating a column accroding to age as age_group

In [25]:
```
labels = ['Young Adult','Adult','Middle Aged','Senior']
df['age_group'] = pd.qcut(df['age'], q = 4, labels = labels)
```

In [31]:
```
df[['age','age_group']].sort_values(by = 'age')
```

Out[31]:

	age	age_group
25	18	Young Adult
2538	18	Young Adult
702	18	Young Adult
1449	18	Young Adult
3506	18	Young Adult
...	...	...
2426	70	Senior
3818	70	Senior
3874	70	Senior
1271	70	Senior
3246	70	Senior

3900 rows × 2 columns

In [ ]:

```

```

Maintaining readability by creating new column 'purchase_freq_days' based ono old column 'frequency_of_purchases'

In [33]:

```
frequency_mapping = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Bi-Weekly' : 14,
    'Monthly' : 30,
    'Quarterly' : 90,
    'Annually' : 365,
    'Every 3 Months' : 90
```

```
}
df['purchase_freq_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

In [35]:
```
df[['frequency_of_purchases','purchase_freq_days']].head(10)
```

Out[35]:

	frequency_of_purchases	purchase_freq_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
5	Weekly	7
6	Quarterly	90
7	Weekly	7
8	Annually	365
9	Quarterly	90

In [ ]:

Data Cleaning

In [37]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[37]:
```
np.True_
```

In [39]:
```
df = df.drop('promo_code_used',axis = 1)
```

In [40]:
```
df.columns
```

Out[40]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_freq_days'],
      dtype='object')
```

In [ ]:

Connecting jupyter to database for analysis

In [41]:
```
!pip install pymysql sqlalchemy
```

```
Collecting pymysql
  Downloading pymysql-1.1.2-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: sqlalchemy in d:\anacondasw\lib\site-packages (2.0.39)
Requirement already satisfied: greenlet!=0.4.17 in d:\anacondasw\lib\site-packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in d:\anacondasw\lib\site-packages (from sqlalchemy) (4.12.2)
Downloading pymysql-1.1.2-py3-none-any.whl (45 kB)
Installing collected packages: pymysql
Successfully installed pymysql-1.1.2
```

In [44]:
```
from sqlalchemy import create_engine

username = "root"
password = "Aditya15"
host = "localhost"
port = "3306"
database = "customer_shopping_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

table_name = "customers"
df.to_sql(table_name, engine, if_exists="replace", index=False)

pd.read_sql("SELECT * FROM customers LIMIT 5;", engine)
```

Out[44]:

	customer_id	age	gender	item_purchased	category	purchase_amount_(usd)	location	size	color	season	review_rating	su
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	

In [ ]: